Cutoff-free positivity of localized Weil forms through the (log 11)/2 threshold

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Abstract

We study the complete geometric Weil form on complex-valued functions supported in a compact interval. We give a computer-assisted certificate of strict positivity in both parity sectors for every support radius $0 < a \le (\log 11)/2$. The proof represents the archimedean form as a diagonal digamma operator plus positive rank-one boundary responses of exponential Green kernels. It retains 350 complete physical responses and all their cross terms with the Fourier complement. Parseval identifies the total cross covariance by finite integrals, and increasing diagonal weights bound its infinite tail. A finite Schur matrix then provides a lower bound for the complete operator, rather than only its principal compression. The bounded prime operator is controlled by an exhaustive four-step enumeration on 200 cells whose endpoints are exact rational combinations of prime logarithms. At 256-bit Arb precision, interval Cholesky and complete seven- and six-dimensional Schur eliminations certify the two sectors. The supplement contains the existing full-run sources and results, explicit regeneration commands, a newly rerun prime enumeration, fresh elimination of the retained small matrices, and exact symbolic Green-kernel checks. The historical large matrix construction was not repeated during submission preparation. The closed-form comparison uses the smooth compact core and preserves the negative odd pole contribution. Block Schur complementation itself is standard; the contribution is the explicit complete-form certificate and its endpoint. This is finite-window positivity, not a proof of all-scale positivity, a new zero-free region, or the Riemann hypothesis.

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1. Introduction

Weil's criterion expresses the Riemann hypothesis as positivity of an explicit quadratic form on every compact support. At a fixed support the form defines a semibounded self-adjoint operator with compact resolvent. Suzuki [1] gives an operator realization, continuity of the lowest eigenvalue with respect to the support radius, and small-window positivity. The question considered here is quantitative: certify positivity for every complex test function in the specific window

$$a_* = \frac{1}{2}\log 11 = 1.198947636399185\ldots .$$

Theorem 1.1. Let A_a be the Friedrichs operator of the geometric Weil form on $(-a, a)$ in the normalization (2.1). Then

$$\inf \sigma(A_a) > 0 \qquad (0 < a \le a_*). \tag{1.1}$$

Both parity restrictions are strictly positive. If the first degeneracy radius is defined by $a_{\deg} = \inf\{a > 0 : \inf \sigma(A_a) \le 0\}$, with the infimum of the empty set taken to be infinity, then $a_{\deg} > a_*$.

The proof uses a diagonal realization of the archimedean part and positive boundary responses of exponential Green kernels. A finite initial response family is retained in the lower operator. Its cross terms with the entire Fourier complement are evaluated by Parseval; a monotone diagonal weight pays the remaining infinite sum. Thus the endpoint assertion concerns the complete closed form, not just a finite matrix compression.

Block Schur complementation, Parseval's identity and monotone tail bounds are standard. The contribution is their explicit implementation for this Weil form, including the signed odd pole term, and the certified endpoint (1.1). We make no priority claim for the abstract Schur lemma below.

Zhu [2] proves fixed-window positivity at half-width 0.8 using a frequency-envelope reduction and Legendre compression. The same paper withdraws an exploratory positivity claim at half-width 1.19 in Section 7. Liu [3] reports computer-assisted bounds at half-widths 1 and 17/16, using source-exact block Schur and tail compensation. Those are related finite-to-infinite mechanisms, and are not subsumed under a claim that Schur-based certification is new. Our endpoint exceeds those stated half-widths, and our proof uses the exponential-response decomposition (3.4) and the full covariance (5.3). This comparison uses the versions identified in the references, checked on 23 October 2026 (China Standard Time); it is not an exhaustive priority claim.

The theta trials of bramble_trace [4] give unconditional upper bounds for fixed-window Rayleigh values. Such upper bounds do not supply positivity. Conversely, our lower certificate neither identifies the ground state nor estimates the sharp minimum. It gives no result at all support radii and does not prove RH or a new zero-free region.

2. The form and its parity restrictions

Use $\widehat{f}(t) = \int_{\mathbb{R}} f(x)e^{-itx}\,dx$ and $C_f(u) = \int_{\mathbb{R}} f(x+u)\overline{f(x)}\,dx$. On smooth compact functions put

$$\begin{aligned}\mathfrak{a}_a[f] &= \frac{1}{2\pi}\int_{\mathbb{R}} h(t)|\widehat{f}(t)|^2\,dt + 2\,\mathrm{Re}\big(\widehat{f}(i/2)\overline{\widehat{f}(-i/2)}\big)\\ &\quad - 2\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\,\mathrm{Re}\,C_f(\log n),\\ h(t) &= \mathrm{Re}\,\psi(1/4+it/2) - \log\pi.\end{aligned} \tag{2.1}$$

Here $\Lambda(p^k) = \log p$ and vanishes otherwise. The sum is finite on each fixed support. We use the closure of this semibounded form. The closed realization, compact resolvent, and continuity of its lowest eigenvalue are the operator-theoretic inputs from [1]. In particular, $C_c^\infty(-a,a)$ is a form core by this realization.

Reflection commutes with the form. Identify each parity space unitarily with $L^2(0,a)$ by $f(x) = v(x)/\sqrt{2}$ for $x > 0$ and $f(-x) = \varepsilon v(x)/\sqrt{2}$, where $\varepsilon = +1$ or -1. The pole term becomes $\varepsilon|\langle p_\varepsilon, v\rangle|^2$, where

$$p_\varepsilon(x) = e^{x/2} + \varepsilon e^{-x/2}. \tag{2.2}$$

In particular $\|p_-\|_2^2 = 2(\sinh a - a)$; this negative term is retained.

Write $d_n = \log n$ and $\alpha_n = \Lambda(n)/\sqrt{n}$. The half-interval prime operator is the finite sum

$$\begin{aligned}(B_{\mathrm{p}}^\varepsilon v)(x) = -\sum_{n\ge 2}\alpha_n\big[&\mathbf{1}_{x+d_n<a}v(x+d_n) + \mathbf{1}_{x>d_n}v(x-d_n)\\ &+ \varepsilon\mathbf{1}_{0<d_n-x<a}v(d_n-x)\big].\end{aligned} \tag{2.3}$$

Endpoint values in these indicators do not matter. At $a = a_*$ the active values are $2,3,4,5,7,8,9$. The $n = 11$ overlap has measure zero.

The orthonormal bases used throughout are

$$\begin{aligned}&e_0^+(x) = a^{-1/2}, && e_q^+(x) = (2/a)^{1/2}\cos(q\pi x/a)\quad (q\ge 1),\\ &e_q^-(x) = (2/a)^{1/2}\sin((q+1/2)\pi x/a)\quad (q\ge 0).\end{aligned} \tag{2.4}$$

Let $k_q^+ = q\pi/a$, $k_q^- = (q+1/2)\pi/a$ and

$$D_\varepsilon e_q^\varepsilon = d_q^\varepsilon e_q^\varepsilon, \qquad d_q^\varepsilon = h(k_q^\varepsilon). \tag{2.5}$$

The symbol grows logarithmically and is bounded below. Thus its diagonal form is closed after a scalar shift. Smooth compact parity tests belong to its form domain, as follows from rapid decay of their coefficients.

3. Positive boundary responses

For $j \ge 0$ set $c_j = 2j + 1/2$ and

$$r_j^\varepsilon(x) = \frac{e^{-c_j a}(e^{c_j x} + \varepsilon e^{-c_j x})}{\sqrt{1-\varepsilon e^{-2c_j a}}}. \tag{3.1}$$

Lemma 3.1. On smooth compact parity tests, the half-interval form is

$$\mathfrak{a}_a^\varepsilon[v] = \langle D_\varepsilon v, v\rangle + \langle B_{\mathrm{p}}^\varepsilon v, v\rangle + \varepsilon|\langle p_\varepsilon, v\rangle|^2 + \sum_{j\ge 0} |\langle r_j^\varepsilon, v\rangle|^2. \tag{3.2}$$

The last series is nonnegative and convergent on these tests.

Proof. The convergent digamma difference expansion is

$$h(t) = h(0) + \sum_{j\ge 0}\left(\frac{2}{c_j} - \frac{2c_j}{c_j^2 + t^2}\right). \tag{3.3}$$

The inverse Fourier transform of $2c/(c^2+t^2)$ is $e^{-c|x|}$. On the half interval, its parity-restricted kernel is $E_c^\varepsilon(x,y) = e^{-c|x-y|} + \varepsilon e^{-c(x+y)}$. Let R_c^+ be the Green operator of $-\partial_x^2 + c^2$ with Neumann conditions at both ends; let R_c^- have Dirichlet condition at zero and Neumann condition at a. Their eigenbases are (2.4). For $x \le y$ their kernels, multiplied by $2c$, are respectively

$$\frac{2\cosh(cx)\cosh(c(a-y))}{\sinh(ca)}, \qquad \frac{2\sinh(cx)\cosh(c(a-y))}{\cosh(ca)}.$$

Subtracting E_c^ε and using symmetry gives the exact identity

$$2cR_c^\varepsilon - E_c^\varepsilon = r_c^\varepsilon \otimes r_c^\varepsilon, \tag{3.4}$$

where r_c^ε is (3.1) with c_j replaced by c. Apply (3.4) termwise to a finite partial sum of (3.3). The diagonal part is the corresponding partial sum of $h(k_q^\varepsilon)$; the difference is the sum of the positive rank-one forms.

The partial sums of $h(t) - h(0)$ increase to that function, so their quadratic forms converge by monotone convergence in both Fourier realizations. For compact tests separated from a, the inner products with (3.1) decay exponentially in c_j. Hence the rank-one series also converges. Adding (2.2) and (2.3) proves (3.2). □

This identity is used on the smooth compact core. The proof below passes to the complete Weil form by closed-form comparison. It does not require identifying the full zero-extension form domain with a periodic Fourier space, nor declaring the infinite response family bounded on L^2.

4. A lower operator on the complete complement

Fix $a = a_*$ and one parity. Define the bounded self-adjoint operator

$$B_0 = B_{\mathrm{p}}^\varepsilon + \sum_{j=0}^{499} r_j^\varepsilon \otimes r_j^\varepsilon + \varepsilon p_\varepsilon \otimes p_\varepsilon, \qquad F = (r_{500}^\varepsilon, \dots, r_{849}^\varepsilon). \tag{4.1}$$

Thus $F : \mathbb{C}^{350} \to L^2(0, a_*)$ is finite rank. Lemma 3.1 gives on the smooth compact core

$$\mathfrak{a}_{a_*}^\varepsilon \ge D_\varepsilon + B_0 + FF^*. \tag{4.2}$$

Let P project onto $e_0^\varepsilon, \dots, e_{399}^\varepsilon$, and $Q = I - P$. The prime bound proved in Section 6 is $\|B_{\mathrm{p}}^\varepsilon\| < \rho_{\mathrm{p}} = 2.931158$. Consequently

$$B_0 + \rho_\varepsilon I \ge 0, \qquad \rho_+ = \rho_{\mathrm{p}}, \qquad \rho_- = \rho_{\mathrm{p}} + 2(\sinh a_* - a_*). \tag{4.3}$$

On the complete infinite-dimensional Q space put

$$Te_q^\varepsilon = t_q e_q^\varepsilon, \qquad t_q = d_q^\varepsilon - \rho_\varepsilon \quad (q \ge 400). \tag{4.4}$$

The summands in (3.3) increase with $t \ge 0$, so the sequence t_q is increasing. Direct interval evaluation verifies $t_{400} > 0$ in both parities. Define

$$B_F = PF, \quad V = QF, \quad M = P(D_\varepsilon + B_0)P + B_F B_F^*, \quad C = PB_0Q + B_F V^*. \tag{4.5}$$

Here $C : QL^2 \to PL^2$ is bounded. On $PL^2 \oplus D(T^{1/2})$ consider

$$\ell[x,y] = \langle Mx, x\rangle + 2\,\mathrm{Re}\langle Cy, x\rangle + \|T^{1/2}y\|^2. \tag{4.6}$$

It is a closed semibounded form, since the cross term is a bounded perturbation and P is finite dimensional. On every smooth compact test, subtracting (4.6) from (4.2) leaves

$$\langle Q(B_0 + \rho_\varepsilon I)Qy, y\rangle + \|V^* y\|^2 \geq 0. \tag{4.7}$$

Therefore $\mathfrak{a}^{\varepsilon}_{a_*} \succeq \ell$ on that core. For completeness, this comparison extends to the target form closure: shift both forms by a sufficiently large scalar to make them nonnegative. A sequence from the smooth core converging in the shifted target form norm is Cauchy in the shifted ℓ norm, by (4.7). Closedness of ℓ places its L^2 limit in $D(\ell)$, and passage to the limit preserves the inequality. Thus

$$\mathfrak{a}^{\varepsilon}_{a_*} \succeq \ell \quad \text{on the complete target form domain.} \tag{4.8}$$

Completing the square in (4.6) gives

$$\ell[x, y] = \|T^{1/2}y + T^{-1/2}C^* x\|^2 + \langle (M - CT^{-1}C^*)x, x\rangle. \tag{4.9}$$

If a finite matrix G satisfies $0 < G \preceq M - CT^{-1}C^*$, this form has a strictly positive uniform lower bound on L^2. Indeed, $T \geq \tau Q$ for $\tau = t_{400} > 0$, and

$$\|y\| \leq \tau^{-1/2}\|T^{1/2}y + T^{-1/2}C^* x\| + \tau^{-1}\|C\|\|x\|.$$

Together with $G \geq gP$ for some $g > 0$, this controls both components by the right side of (4.9). No numerical spectral-gap value is inferred from an individual elimination pivot.

5. Parseval covariance and the weighted tail

For $q \geq 400$ let

$$c_q = PB_0 e^{\varepsilon}_q, \quad v_q = F^* e^{\varepsilon}_q, \quad U_q = c_q + B_F v_q = C e^{\varepsilon}_q.$$

Since P is finite dimensional and C is bounded, Parseval gives a trace-norm-convergent positive matrix series

$$R_{\mathrm{tot}} = CC^* = \sum_{q \geq 400} U_q U_q^*. \tag{5.1}$$

This total covariance is evaluated without truncating the q sum. Put

$$\begin{aligned} R_0 &= PB_0^2 P - (PB_0 P)^2, \\ X &= PB_0 F - (PB_0 P)B_F, \qquad G_Q = F^* F - B_F^* B_F. \end{aligned} \tag{5.2}$$

Expanding CC^* yields

$$R_{\mathrm{tot}} = R_0 + XB_F^* + B_F X^* + B_F G_Q B_F^*. \tag{5.3}$$

All atom–atom and atom–base cross terms occur in this identity. The matrices on the right involve finite families of elementary exponentials and partial translations or reflections. Their integrals are evaluated exactly in terms of exponentials, with outward interval enclosures of the resulting constants.

With $q_f = 8000$ define

$$R_1 = \sum_{400 \leq q < q_f} U_q U_q^*, \qquad G = M - \sum_{400 \leq q < q_f} \frac{U_q U_q^*}{t_q} - \frac{R_{\mathrm{tot}} - R_1}{t_{q_f}}. \tag{5.4}$$

Monotonicity of t_q and positivity of each matrix summand imply

$$\sum_{q \geq q_f} \frac{U_q U_q^*}{t_q} \preceq \frac{R_{\mathrm{tot}} - R_1}{t_{q_f}}, \qquad G \preceq M - CT^{-1}C^*. \tag{5.5}$$

Thus a positive-definiteness certificate for this 400×400 matrix proves positivity of the complete lower form (4.6), and hence of the complete Weil form. The finite family $500 \leq j < 850$ is sufficient for this lower certificate; the omitted physical responses are nonnegative in (3.2). The Fourier complement in (4.4) and (5.1) remains infinite.

6. The prime operator bound

Let $\mathcal{T}$ be (2.3) with every coefficient replaced by its positive absolute weight and without its leading minus sign. It is self-adjoint, preserves nonnegative functions, and dominates both signed parity operators pointwise in absolute value. Schur's test gives

$$\|B_{\mathrm{p}}^{\varepsilon}\| \le \|\mathcal{T}\|, \qquad \|\mathcal{T}\|^4 = \|\mathcal{T}^4\| \le \operatorname*{ess\,sup}_{0<x<a_*} (\mathcal{T}^4\mathbf{1})(x). \tag{6.1}$$

Each edge is a partial translation $x \mapsto x \pm \log n$ or reflection $x \mapsto \log n - x$, with the support specified in (2.3). The four-step row sum is constant on cells determined by their support endpoints and inverse images under words of length at most three. Represent every endpoint by its rational coefficient vector in $\log 2, \log 3, \log 5, \log 7, \log 11, \log 13$. Equality of these vectors is exact by unique factorization. Arb certifies the strict ordering of distinct endpoints. Thus cell enumeration is exhaustive rather than a sampled grid.

The supplied enumeration has 201 endpoints and 200 open cells. Within each cell it computes the four-step row sum as a polynomial in the positive edge weights $\Lambda(n)/\sqrt{n}$. Its maximum is enclosed by

$$\operatorname{ess\,sup} \mathcal{T}^4\mathbf{1} \in [73.817062678870728252782186080550472646147554246990346635328981273454381577 \pm 2.16\times10^{-73}]. \tag{6.2}$$

Taking the positive fourth root gives $2.931157735700846397899367728988589956 28\ldots < 2.931158$, which proves (4.3). Values at the finite set of endpoints do not affect the operator norm. The enumeration source and its output are included in the supplement as `certify_log11_prime_four_step_degree.py` and `prime-graph-check.txt`.

7. Endpoint certificates and proof of Theorem 1.1

The certificate uses $K = 400$, an initial response count 500, the retained family $500 \le j < 850$, $q_f = 8000$, and 256-bit Arb arithmetic. A preliminary $R = 280$ construction selects exact dyadic Householder vectors; the resulting congruences act on the first 280 coordinates and leave the remaining 120 coordinates unchanged. Floating-point linear algebra selects these vectors only. Their normalization and all subsequent congruences are verified by interval arithmetic.

The matrix (5.4) is checked after that orthogonal congruence. Its block sizes are $7 + 393$ in the even sector and $6 + 394$ in the odd sector. Interval Cholesky verifies positivity of the larger block; a verified solve forms the remaining Schur complement; symmetric pivoting and interval elimination certify the complete smaller block.

The relevant positive intervals are:

	+	−
minimum complement pivot	$0.040799562910896252229291041959 \pm 10^{-30}$	$0.007336690629086577841762831004 \pm 10^{-30}$
last core pivot	$3.68238777398334030\,10^{-50} \pm 5.91\,10^{-68}$	$3.643502827335460523 88\,10^{-45} \pm 7.93\,10^{-66}$

(7.1)

The full core permutations are $(6, 2, 3, 4, 5, 1, 0)$ and $(3, 5, 4, 1, 0, 2)$, respectively. All seven even pivots and all six odd pivots have strictly positive lower endpoints. The full arrays and the evaluated tail quantities t_{400} and t_{8000} are in the two supplied result files. The checker rejects a nonpositive or indeterminate pivot and an incomplete pivot list.

To identify this computation with (5.4), the source variables are `BF = PF`, `RP = R_0`, `XF = X`, `GQ = G_Q`, and `RTOT = R_tot` after the same ordered Householder congruence. The loaded scalar-complement base is `P0 = P(D+B0)P - RP/t400`. Accordingly the source expression

```
G = P0 + RP/t400 + BF*BF.transpose() - G1 - (RTOT-R1)/tQF
```

is precisely (5.4). `G1` is its finite weighted prefix. This algebraic identification is essential: a positive result for another compressed model would not prove (1.1).

The intervals (7.1), together with the complete positive pivot lists, prove $G > 0$ in each parity. Equations (4.8), (4.9) and (5.5) therefore give a positive lower bound for A_{a_*}.

For $a < a_*$, zero extension embeds smooth tests in the larger window without changing (2.1). The min–max principle, extended through the form closures, gives $\inf\sigma(A_a) \ge \inf\sigma(A_{a_*}) > 0$. This proves (1.1). Continuity of the lowest eigenvalue in [1] extends positivity to a neighborhood to the right of a_*, proving the stated strict inequality for a_{deg}. □

8. Reproduction and limitations

The supplement includes the existing endpoint certificate, its six construction sources, all stage logs, the two interval result files, and the frozen-result verifier. It also includes the four-step prime enumeration and a symbolic check of (3.4).

The historical complete matrix construction was performed on 10 September 2026. For this submission, the supplied metadata, complete pivot counts, strict interval signs and frozen-file bindings were checked again, and the stored small Schur matrices were eliminated again with Arb. The grid-free prime enumeration and symbolic identities were also rerun. The large parent matrices were not regenerated during this preparation. Checking frozen results alone does not independently establish their generation; the complete regeneration commands and sources are supplied so that the calculation can be reproduced without private files.

The lightweight command is

```
uv run --with numpy python certificate/verify_frozen_results.py
```

The other short checks are

```
uv run --with python-flint python certify_log11_prime_four_step_degree.py 256 11
uv run --with sympy python verify_green_response_identities.py
```

`REPRODUCTION.md` gives the full sequential construction from an empty working directory. Its dependencies are NumPy and python-flint; SymPy is used only for the extra algebraic check. The complete computation involves repeated interval matrix assembly and is substantially slower than the lightweight checks. In the retained even run, the sharp-prefix stage alone reached $q_f = 8000$ after approximately 1474 seconds.

The result is a strict finite-window lower bound. It does not establish positivity above this window, a sharp spectral gap, all-scale Weil positivity, or RH. The small positive elimination pivots in (7.1) are not eigenvalue bounds. Increasing a finite response family has no automatic implication for larger support radii.

References

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[3] V. Liu. Certified Weil Positivity Beyond the Unit Window: Source-Exact Block-Schur and Tail-Compensation Bounds for the Riemann Zeta Function. Submitted manuscript (2026). Author's public record: https://www.alphaxiv.org/abs/2609.weil-positivity-riemann-zeta-bounds Frozen source and manuscript, commit b6cd2183c1e79c6c27a34267812a7b2d73ed1b59: https://github.com/luciferyu666/certified-weil-positivity/tree/b6cd2183c1e79c6c27a34267812a7b2d73ed1b59

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